Lists and indexing

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Lists and indexing before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Loops and trace tables

Without opening the old lesson, explain one key idea from Loops and trace tables. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Boolean conditions

Without opening the old lesson, explain one key idea from Boolean conditions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

From a paper algorithm to Python

Without opening the old lesson, explain one key idea from From a paper algorithm to Python. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Lists and indexing trace and record the first step you can explain without help.

Use W2: Trace an update for **List**s and **index**ing. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Lists and indexing explanation against the supplied facts before discussing it.

Use the worked case for **List**s and **index**ing. Proposed explanation: "**Length** is the last nonnegative **index**". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Distinguish a value from an index

> Predict the three **output**s. If the second print uses values[3] instead, where does execution stop and which later **output** is not produced?

```
values = [4, 9, 2]
print(values[1])
print(values[-1])
print(len(values))
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Separate index and value

> Separate index and value. Apply the supplied Lists and indexing facts and keep your reasoning.

For counts = [4, 7, 4], give the **length**, valid nonnegative indices and value at **index** 1. Explain how the two **item**s with value 4 differ.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace an update

> Trace an update. Apply the supplied Lists and indexing facts and keep your reasoning.

Start with counts = [4, 7, 4]. Run counts[0] = 6. Write the resulting **list**, its **length** and the value at **index** 2. Predict before running.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Repair a position mistake

> Repair a position mistake. Apply the supplied Lists and indexing facts and keep your reasoning.

A learner loops with for count in counts and tries to read counts[count] for counts = [4, 7, 4]. Identify the first attempted **index** and error. Explain how to print each value directly.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Process new data independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Process new data independently. Apply the supplied Lists and indexing facts and keep your reasoning.

Start with readings = [3, 0, 3, 2]. Replace the second **item** with 5 using its **index**. Write a loop that adds all updated readings and prints one final total. Trace every update of the total.

Then test the summing part alone on [] and [6].

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Keep the information you need

> Keep the information you need. Apply the supplied Lists and indexing facts and keep your reasoning.

Compare [1, 5, 2] and [5, 1, 2]. State their totals and first **item**s. Would storing only the total answer which first **item** was larger? Explain.

> **My reasoning and evidence**

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Lists and indexing to Python readiness portfolio, identify a limit and choose the next useful test.

Add a labelled **list**s and **index**ing **evidence** sheet to Python readiness portfolio. Show how you can distinguish a **list index** from the value stored at that position.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Lists and indexing

> Indexes are positions, not values. A list of shelf counts is read by position. Explain the labelled step before checking the changed condition.

Indexes are positions, not values. A **list** of shelf counts is read by position.

Python 3: counts=[6,1,8]; print(counts[0]); print(counts[-1]); print(len(counts)).

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Zero-based list positions



Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / PYTHON-BRIDGE/7

Lists and indexing

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Draw positions

Indexes 0,1,2 contain 6,1,8 respectively.

↓

2. Read two positions

Index 0 returns 6; index -1 returns the final value 8.

↓

3. Check length boundary

Length is 3, but index 3 is beyond the final valid positive index 2.

CHANGE ONE THING

Replace the second print with print(counts[3]).

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Label the three **index** positions. Predict all original **output**s, then name the first **output** and stopping point after using **index** 3.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation.

Use actual Python 3 results to check the **prediction** after reasoning first.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Python readiness portfolio record.

Project: Python readiness portfolio

Current milestone: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Indexes are positions, not values. A **list** of shelf counts is read by position.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? **Length** is 3, but **index** 3 is beyond the final valid positive **index** 2.

1. Read two positions

2. Check **length** boundary

3. Draw positions

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check **length** boundary **Length** is 3, but **index** 3 is beyond the final valid positive **index** 2. **List** positions begin at zero.

A negative **index** counts from the end; the **list length** counts entries and is not itself a valid positive **index** for the final entry.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Replace the second print with print(counts[3]). Which conclusion is supported by the supplied facts?

1. The first print shows 6, then **IndexError** stops the script before the **length** print.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The first print shows 6, then **IndexError** stops the script before the **length** print. The attempted position does not exist.

Repair a missing building block

Review Loops and trace tables (python-bridge/6). Goal: Trace variable values before and after each loop iteration. Short practice: **List** the four visited values and their running totals.

Explain why 5 is absent and what prints for range(1,1).. Return to this week and explain what you changed.

Review Boolean conditions (python-bridge/4). Goal: Translate stated requirements into comparisons. Short practice: Give both comparison results for ages 11, 15 and 16.

Which comparison rejects each outside value, and why does and matter?. Return to this week and explain what you changed.

Review From a paper algorithm to Python (python-bridge/1). Goal: Translate a short paper calculation into a Python script. Short practice: Match each paper quantity to its Python name.

Why does the spare count get added once, and what prints when spare is five?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | |
> |---|
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> | |
> | |
> | |

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Lists and indexing picture from its labels, then record a relevant limit.

01. Lists and indexing: the case

AI ACADEMY / PYTHON-BRIDGE/7 · 01 · LOOK AT THE FACTS

Lists and indexing: the case

The situation

Indexes are positions, not values. A list of shelf counts is read by position.

Supplied evidence

Python 3: counts=[6,1,8]; print(counts[0]); print(counts[-1]); print(len(counts)).

Trace lists and indexing and explain the changed program before running it.

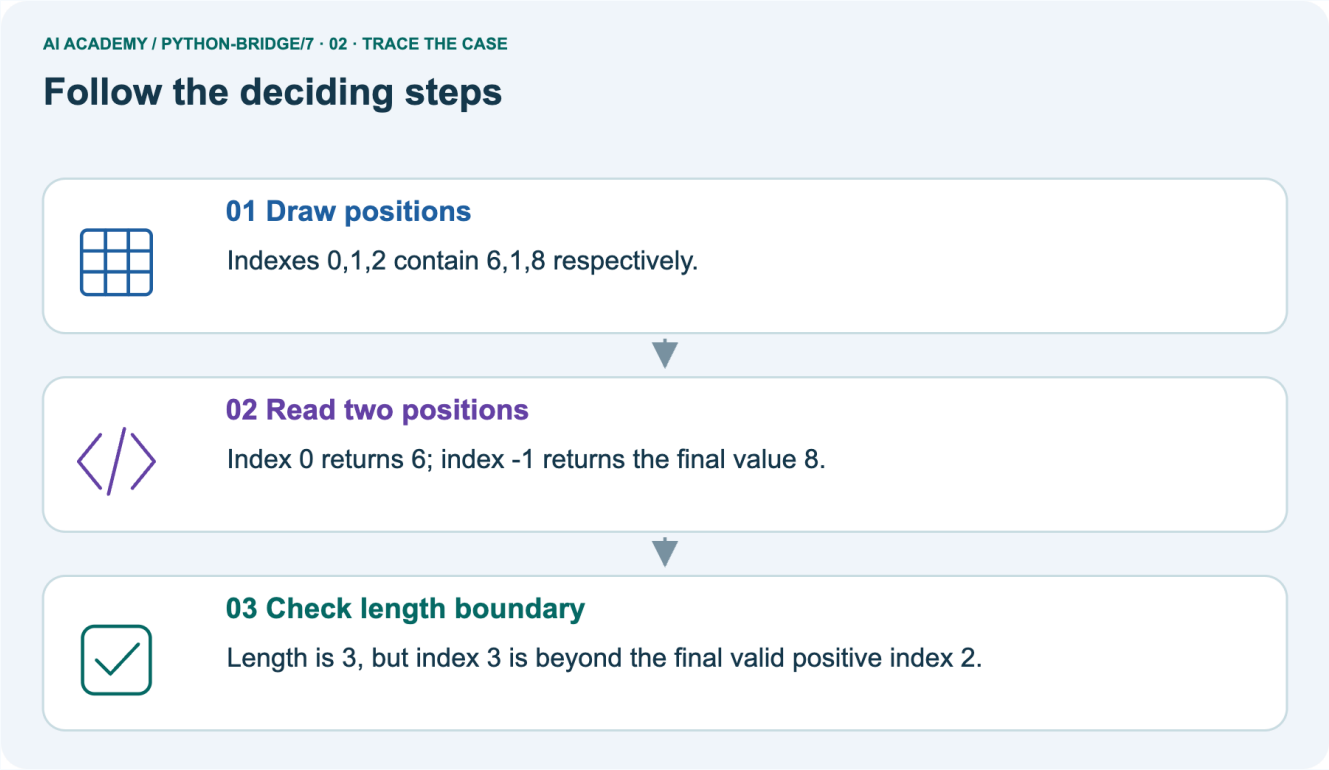
Follow the labelled picture with the matching explanation. Open the vector diagram

Indexes are positions, not values. A **list** of shelf counts is read by position. Supplied facts: Python 3: counts=[6,1,8]; print(counts[0]); print(counts[-1]); print(len(counts)).

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Draw positions: **Index**es 0,1,2 contain 6,1,8 respectively. → Read two positions: **Index** 0 returns 6; **index** -1 returns the final value 8. → Check **length** boundary: **Length** is 3, but **index** 3 is beyond the final valid positive **index** 2.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Zero-based list positions



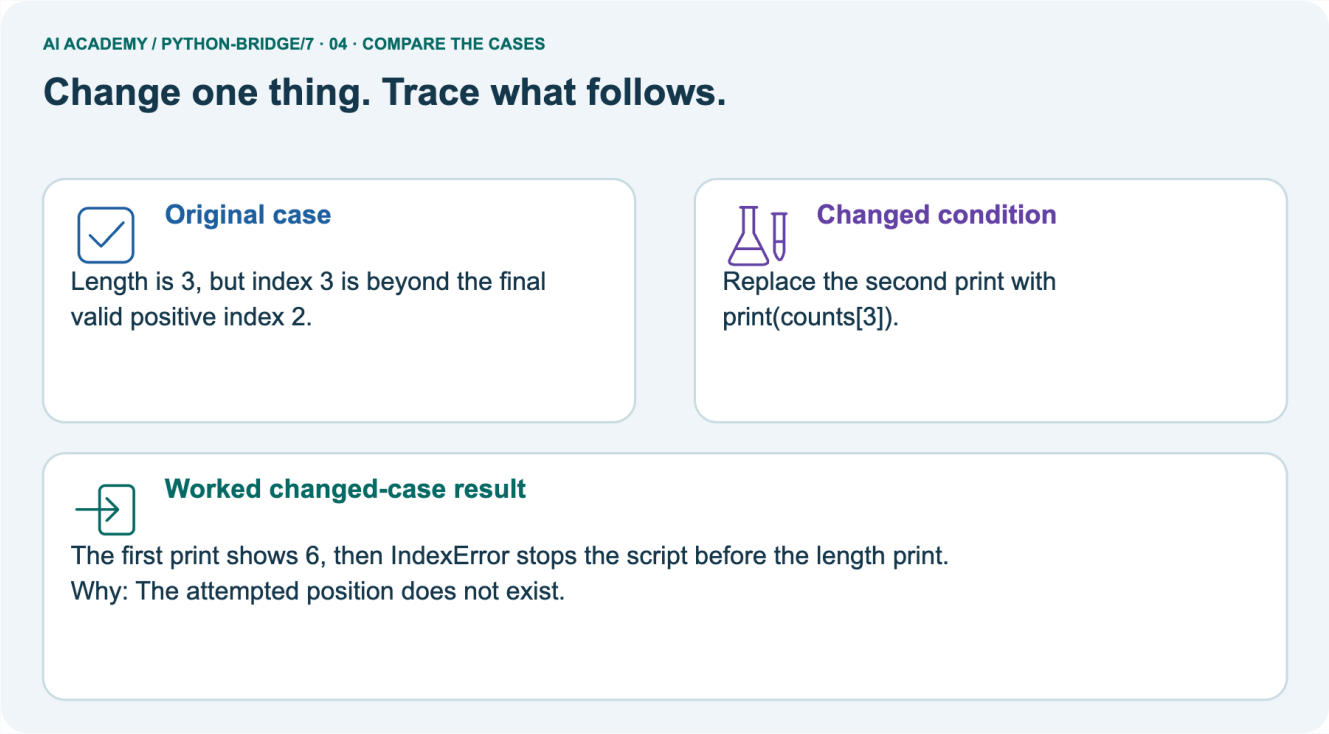
Follow the labelled picture with the matching explanation. Open the vector diagram

Zero-based **list** positions. Stored counts: **Index** 0 = 6, **Index** 1 = 1, **Index** 2 = 8.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

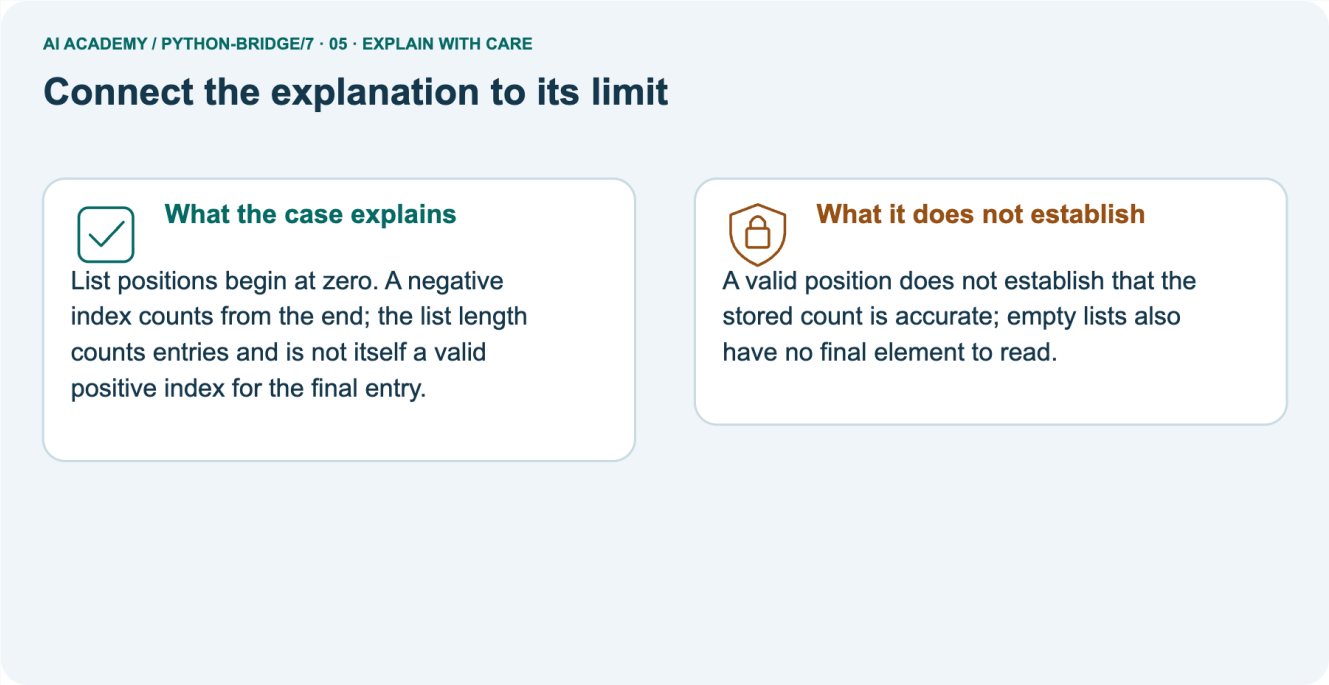
Original case: **Length** is 3, but **index** 3 is beyond the final valid positive **index** 2. Changed condition: Replace the second print with print(counts[3]).

Expected result: The first print shows 6, then **IndexError** stops the script before the **length** print. Reason: The attempted position does not exist.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

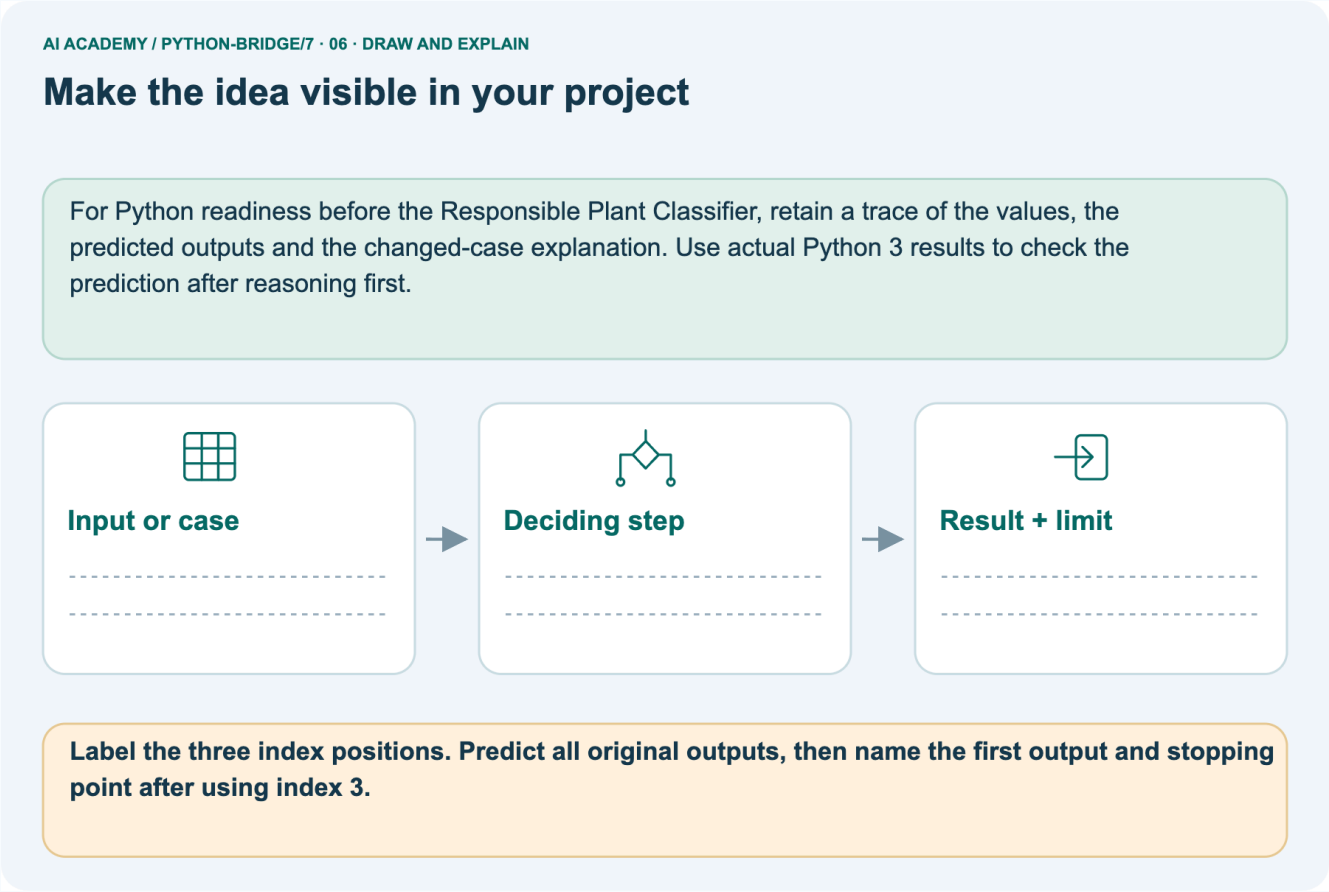
List positions begin at zero. A negative **index** counts from the end; the **list length** counts entries and is not itself a valid positive **index** for the final entry.

Limit: A valid position does not establish that the stored count is accurate; empty **list**s also have no final element to read.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Label the three **index** positions. Predict all original **output**s, then name the first **output** and stopping point after using **index** 3.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.